

Introduction: Evidence Gaps in a Rapidly Expanding Field

The Free Energy Principle and Active Inference Framework

I

The Free Energy Principle and Active Inference Framework

II

The Free Energy Principle (FEP), introduced by Karl Friston, proposes that self-organizing systems maintain their structural and functional integrity by minimizing variational free energy—an upper bound on sensory surprise [friston2006free, friston2010free]. Under this principle, living systems are cast as approximate Bayesian inference engines that build generative models of their environment and act to reduce the discrepancy between predicted and observed states. Active Inference (AIF) extends this picture from passive perception to goal-directed behavior: agents select actions that bring about observations consistent with their preferred states, unifying perception, learning, and decision-making within a single variational framework [parr2022active, friston2017active]. Since its initial formulation for sensorimotor control, AIF has been applied to navigation, visual foraging, language comprehension, social cognition, and multi-agent coordination. Bayesian mechanics [sakthivadivel2023bayesian] has further strengthened the mathematical foundations of the FEP by grounding Markov

Challenges Posed by Rapid Literature Growth I

Challenges Posed by Rapid Literature Growth II

The active inference literature has grown at a compound annual rate of 20.36% across 2005–2026, with annual output accelerating sharply after 2013. While early research concentrated on theoretical neuroscience, the field has since diversified across biology (C5), robotics (C2), computational psychiatry (C4), algorithm scaling (B), and formal mathematics (A1). With $N = 817$ papers spanning 8 categories across 3 domains, no prior automated system tracks hypothesis-level evidence across the full corpus. This creates three interrelated challenges. First, the balance of evidence for core claims—such as FEP universality or the physical realism of Markov blankets—cannot be assessed without structured, hypothesis-specific extraction at corpus scale. Second, because the relationship between mathematical formalisms and empirical evidence is frequently implicit, systematic evidence synthesis demands substantial manual effort: Knight et al. [knight2022fep] required human annotators to manually code hundreds of papers. Third, new entrants must navigate a literature weighted toward broad qualitative philosophy (A2), interspersed with specialized applied subfields whose findings are difficult to

Related Work and Prior Meta-Analyses I

Several prior efforts have surveyed aspects of the Active Inference landscape. Sajid et al. [sajid2021active] compare active inference with alternative decision-making frameworks; Da Costa et al. [dacosta2020active] synthesize the discrete-state-space formulation; Lanillos et al. [lanillos2021active] survey robotics applications; Smith et al. [smith2021computational] provide a tutorial bridging theory and empirical data; and Millidge et al. [millidge2021understanding] examine information-theoretic foundations of exploration behavior. Ramstead et al. [ramstead2018answering] extend the FEP to questions of biological self-organization, while Pezzulo et al. [pezzulo2015active] connect active inference to homeostatic regulation. Millidge [millidge2024retrospective] provides a practitioner's retrospective confirming that AIF's strongest demonstrated results arise from novel discrete generative models, while scalability relative to deep reinforcement learning remains the field's central open challenge.

Related Work and Prior Meta-Analyses II

Parallel to these synthesis efforts, Sanjeev V. Namjoshi's 2026 textbook, *Fundamentals of Active Inference* [namjoshi2026fundamentals], provides a comprehensive, computationally explicit foundation for the field designed for engineers. In conjunction with this text, Namjoshi developed the aif-fep-db repository [namjoshi2026aiffepdb]—an open-source, dynamically updated database of scraped and tagged publications covering active inference, the free energy principle, and predictive processing. While aif-fep-db curates and categorizes the literature to facilitate reproducible systematic reviews and interactive Dash-based exploration, it functions primarily as a modular bibliographic foundation rather than an automated hypothesis evaluation engine.

Related Work and Prior Meta-Analyses III

Closest to our work, Knight, Cordes, and Friedman [knight2022fep] conducted a systematic literature analysis of publications using the terms “Free Energy Principle” or “Active Inference,” with an emphasis on works by Karl J. Friston. Their analysis—maintained by the Active Inference Institute—combined manual annotation of structural, visual, and mathematical features with automated analyses using the Active Inference Ontology at the scale of thousands of citations and hundreds of annotated papers. That study identified six development directions—including broader scope, richer annotation, and transferable approaches—and represents an important precursor to automated meta-analysis of this field.

Related Work and Prior Meta-Analyses IV

These prior works differ from the present study along four dimensions. First, **scale**: narrative reviews cover tens to low hundreds of papers; our pipeline processes $N = 817$. Second, **structure**: prior reviews produce prose summaries rather than machine-queryable knowledge graphs with typed relationships. Third, **temporal tracking**: no prior system computes how evidence for specific hypotheses evolves year over year. Fourth, **automation**: the systematic analysis of Knight et al. [knight2022fep] pioneered quantitative literature analysis but relied on manual annotation, limiting update frequency. Our framework advances this line of work by (1) fully automating assertion extraction via LLM-based hypothesis scoring, (2) constructing a structured, RDF-compatible knowledge graph scored by citation-weighted evidence, and (3) tracking how evidence for core claims evolves over time through temporal trend analysis.

Synergizing Knowledge Graphs and LLMs I

Broadening this synthesis, recent systematic literature initiatives underscore a powerful reciprocal synergy between Large Language Models (LLMs) and Knowledge Graphs: LLMs parse unstructured text to rapidly extract semantic claims, efficiently populating the structured, queryable architecture of the graph [quevedo2025combining, li2024unifying]. We adopt the *nanopublication* [groth2010anatomy]—a minimal, machine-readable unit of scientific evidence comprising a core assertion bound to explicit provenance metadata—as the fundamental serialization format for this extracted knowledge.

This Study: Approach and Overview I

This paper presents a computational meta-analysis of the Active Inference literature ($N = 817$). Rather than relying exclusively on bibliometric metadata or slow manual coding, we deploy a Large Language Model (LLM) to “read” each paper’s abstract and assess its relationship to eight core hypotheses within the FEP paradigm. We serialize these assessments as nanopublications—each encoding an assertion (“Paper X supports Hypothesis Y”) coupled with the LLM’s natural-language reasoning and confidence score. The resulting knowledge graph aggregates these nanopublications and links them to paper metadata, citation networks, subfield classifications, and hypothesis definitions. A citation-weighted scoring formula quantifies the net evidence for or against each hypothesis, producing scores in $[-1, 1]$ that reflect both the direction and strength of published evidence. Importantly, this represents an open-source introductory analysis which will be augmented and extended, and stewarded in collaborative development by the Active Inference Institute (activeinference.org).

Research Questions I

This meta-analysis addresses four primary research questions:

1. **RQ1 (Field Structure):** What is the disciplinary structure and growth trajectory of the Active Inference literature, and how are papers distributed across the three domains—Core Theory (A), Tools & Translation (B), and Application Domains (C)? We expect Domain A to dominate but anticipate measurable diversification into applied domains.
2. **RQ2 (Growth Dynamics):** What are the temporal growth dynamics of the field, and which subfields are experiencing the most rapid expansion? Prior reviews suggest accelerating growth post-2013; we quantify this trajectory and identify which application domains drive it.

Research Questions II

3. **RQ3 (Hypothesis Evidence):** What is the current balance of evidence for and against the eight standard hypotheses, and how has this balance evolved over time? We expect well-established hypotheses (H4 Predictive Coding) to show consensus while philosophically contested claims (H3 Markov Blanket Realism) show mixed evidence. (See hypothesis dashboard and assertion figures in the hypothesis results.)
4. **RQ4 (Tooling Readiness):** What is the state of software tooling and infrastructure for Active Inference research, and what gaps remain? We survey available implementations to identify whether the ecosystem is fragmented or converging.

Scope and Delimitations I

Scope and Delimitations II

This study focuses on the English-language peer-reviewed and preprint literature retrievable from arXiv, Semantic Scholar, and OpenAlex. Our search scope begins at 2005—chosen to capture Energy-Based Model and variational Bayesian antecedents (Helmholtz machines, VAEs, early Bayesian brain formulations [dayan1995helmholtz, lecun2006tutorial]) that share deep mathematical foundations with variational free energy minimization and predated the Free Energy Principle label introduced in 2006 [friston2006free]. The scope includes both the core Active Inference and Free Energy Principle literature and adjacent Energy-Based Model research where it intersects with variational inference or generative modeling—capturing the growing convergence between these traditions. We do not include book chapters or monographs not indexed by these sources, software documentation, or non-English publications. Domain classification uses keyword matching (200+ indicators across 8 categories) rather than expert annotation—a deliberate trade-off favoring reproducibility over precision, whose consequences we quantify in the field overview. Hypothesis scoring relies on LLM-extracted

Principal Contributions I

This work makes five contributions:

1. **A multi-source retrieval and deduplication pipeline** for Active Inference literature, using a canonical identifier hierarchy across three academic databases.
2. **A nanopublication-based knowledge graph schema** encoding directed, confidence-scored assertions about eight core hypotheses with full provenance tracking.
3. **A quantitative field overview** characterizing the growth, domain distribution (A/B/C taxonomy), citation topology, and latent topic structure of the Active Inference literature, with specific attention to how recent benchmark results (detailed in the domain analyses) are reshaping the scalability and application landscape.

Principal Contributions II

4. **An LLM-based hypothesis scoring dashboard** that produces differentiated evidence profiles with temporal trend visualization.
5. **A tooling assessment** of the software ecosystem supporting Active Inference research, including the implemented extraction pipeline, existing software (pymdp, SPM, RxInfer.jl), and knowledge graph infrastructure.

Principal Contributions III

The remainder of this paper is organized as follows. The methodology section describes the five-stage pipeline—the central contribution enabling reproducible, automated evidence synthesis—with separate treatments of literature retrieval, LLM-based assertion extraction, bibliometric analysis, the nanopublication-based knowledge graph, and visualization and variable injection. The hypothesis evidence landscape presents quantitative scoring results (RQ3), followed by the field overview with domain-level analysis (RQ1, RQ2), detailed domain analyses, text analytics, and citation network topology. The conclusion addresses limitations and future directions; the discussion provides community recommendations and open questions. Appendix ?? collects mathematical and algorithmic details; Appendix ?? surveys the tooling landscape (RQ4).